PH7002: Concepts in Statistical Mechanics
§4 Landau Theory

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§4.1 Aperitifs

§4.2 Broken symmetry

§4.3 Order parameter

Example 4.1: Topological defects in nematic liquid crystals
Example 4.2: XY defects

§4.4 Landau theory

§4.5 Continuous phase transition

§4.6 First order phase transitions

§4.7 Inhomogeneous system

§4.8 Validity of mean field theory

Example 4.3: Symmetries and wave equations
Example 4.4: Superfluid order and vortices

§4.9 Extra Notes

Reading

Yeomans sections 4.2, 4.3, 4.4, 4.5 and Sethna chapter 9, sections 10.1 and 10.2.

Mean field theory (MFT) gave the so-called classical values for the critical exponents by expanding the equation of state near the critical point. The expansion parameter was the spontaneous magnetization in the case of the ferromagnet and is an example of the concept of an order parameter.

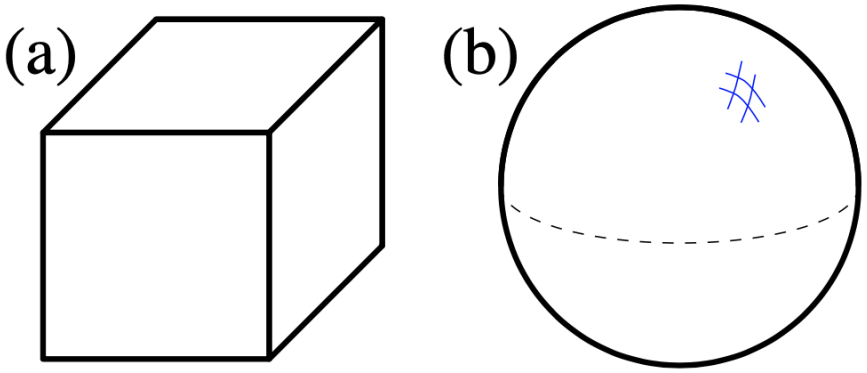
Important themes

1. Any thermodynamic variable that is zero in the un-ordered phase and non-zero in an adjacent (on the phase diagram), usually ordered phase, is a possible choice for an order parameter.
2. Landau free energy, $\mathcal{L}$, depends on the coupling constants K_i and the order parameter ϕ. $\mathcal{L}$ has the property that the state of the system is specified by the absolute minimum (i.e. global) of $\mathcal{L}$ with respect to ϕ.
3. Use Landau theory to describe first order and second order phase transitions.

Broken symmetry

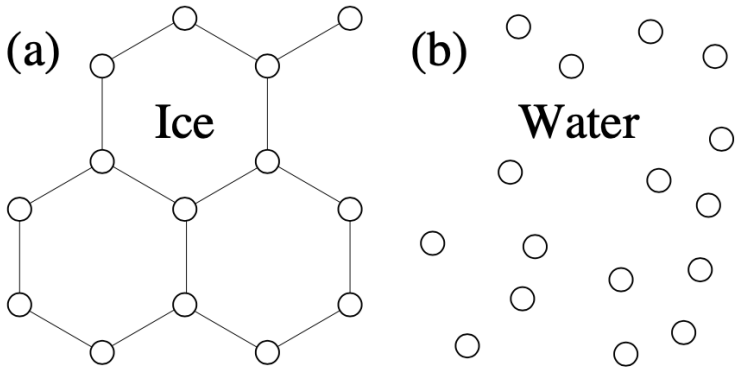
Q: *Consider a cube and a sphere. Which is more symmetric?*

Clearly, the sphere has many more symmetries than the cube. One can rotate the cube by 90° in various directions and not change its appearance, but one can rotate the sphere by any angle and keep it unchanged.



Q: *Which state is more symmetric here?*

Naively, the ice looks much more symmetric; regular arrangements of atoms forming a lattice structure. Ice has a discrete rotational symmetry. It also has a discrete translational symmetry: it is easy to tell if the picture is shifted sideways, unless one shifts by a whole number of lattice units. The water looks irregular and disorganized. On the other hand, if one rotated water by an arbitrary angle, it would still look like water! Water is not a snapshot; it is better to think of it as a combination (or ensemble) of all possible snapshots. While the snapshot of the water shown in the figure has no symmetries, water as a phase has complete rotational and translational symmetry.



Order parameter and phase

This is not to say that different phases always differ by symmetries! Liquids and gases have the same symmetry, and some fluctuating phases in low dimensions do not break a symmetry.

It is safe to say, though, that if the two materials have different symmetries, they are different phases.

Order parameter

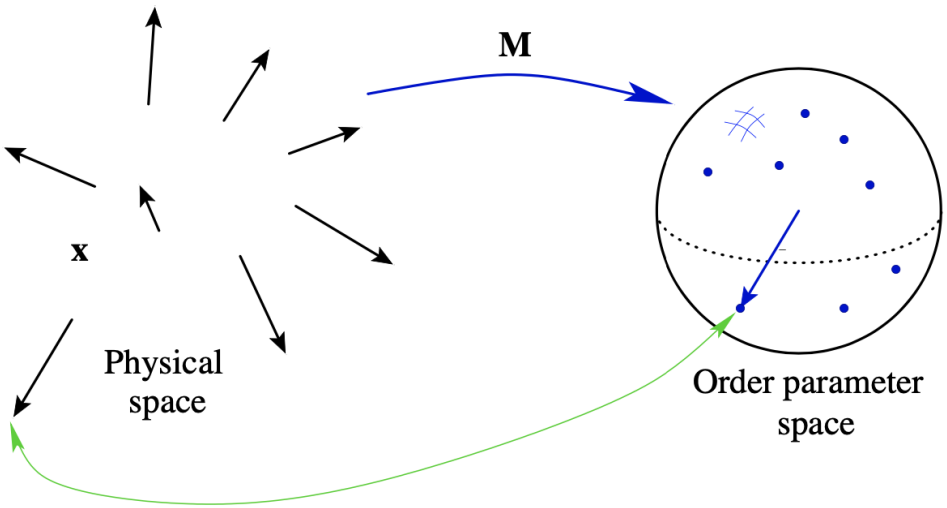
The **order parameter** can be a vector, a tensor, a pseudo-scalar, or a group element of a symmetry group such as SU(N) etc. The order parameter for a given system is not unique; any thermodynamic variable that is zero in the un-ordered phase and non-zero in an adjacent (on the phase diagram), usually ordered phase, is a possible choice for an order parameter.

In a magnet, the important variable is the local direction of the magnetization (an arrow pointing to the 'north' end of the local magnet). The local magnetization comes from complicated interactions between the electrons, and is partly due to the little magnets attached to each electron and partly due to the way the electrons dance around in the material; these details are for many purposes unimportant. The important variables are combined into an **order parameter field** $\phi(\mathbf{x})$.

Magnets

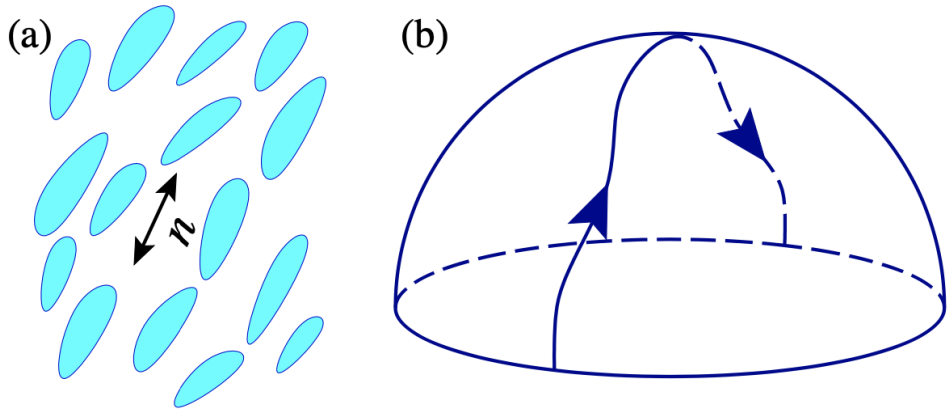
The order parameter field for a magnet is the magnetization $\mathbf{M}(\mathbf{x})$. At each position $\mathbf{x} = (x, y, z)$ we have a direction for the local magnetization $\mathbf{M}(\mathbf{x})$. The length of $\mathbf{M}$ is pretty much fixed by the material, but the direction of the magnetization is undetermined. By becoming a magnet, this material has broken the rotational symmetry. The order parameter $\mathbf{M}$ labels which of the various broken symmetry directions the material has chosen.

The order parameter field $\mathbf{M}(\mathbf{x})$ can be usefully visualized in two different ways. On the one hand, one can think of a little vector attached to each point in space. On the other hand, we can think of it as a mapping from real space into order parameter space. That is, $\mathbf{M}$ is a function which takes different points in the magnet onto the surface of a sphere $\mathbb{S}^2$.



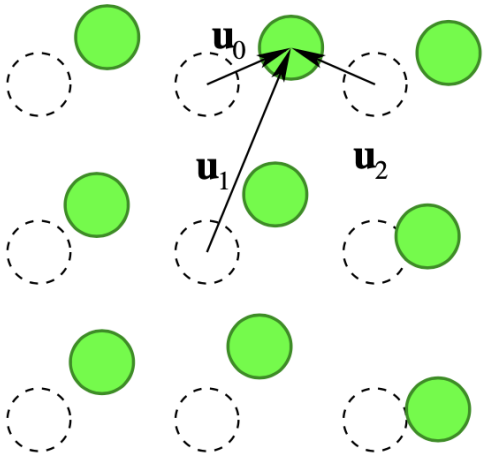
Nematic Liquid Crystals

The liquid crystals in LCD displays (like those in old digital watches) are nematics. **Nematics** are made of long, thin molecules which tend to line up so that their long axes are parallel. Nematic liquid crystals, like magnets, break the rotational symmetry. Unlike magnets, though, the main interaction is not to line up the north poles, but to line up the axes. (Think of the molecules as American footballs, the same up and down.) Thus the order parameter is not a vector $\mathbf{M}$ but a headless vector $\vec{n} = -\vec{n}$. The order parameter space is a hemisphere, with opposing points along the equator identified. This space is called the projective plane, $\mathbb{R}P^2$, by the mathematicians for obscure reasons.



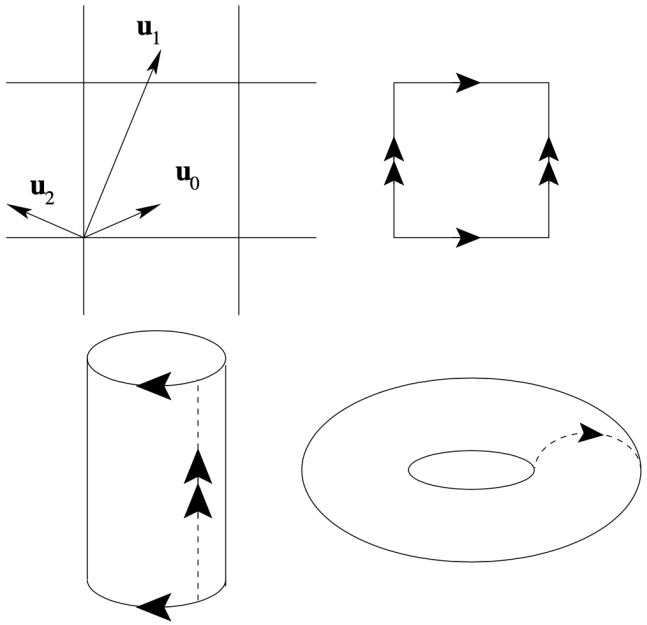
Crystals

For a crystal, the important degrees of freedom are associated with the broken translational order. Consider a two-dimensional crystal which has lowest energy when in a square lattice, but which is deformed away from that configuration. This deformation is described by an arrow connecting the undeformed ideal lattice points with the actual positions of the atoms. If we are a bit more careful, we say that $\mathbf{u}(\mathbf{x})$ is also defined between the lattice positions; there still is a best displacement which locally lines up the two lattices.



The order parameter $\mathbf{u}$ is not really a vector; there is a subtlety. In general, which ideal atom you associate with a given real one is ambiguous. The displacement vector $\mathbf{u}$ can change by a multiple of the lattice constant a when we choose a different reference atom:

The set of distinct order parameters forms a square with periodic boundary conditions. A square with periodic boundary conditions has the same topology as a torus, $\mathbb{T}^2$.



Finding the right order parameter

Guessing the order parameter (or the broken symmetry) is not always so straightforward. For example, it took many years before anyone figured out that the order parameter for superconductors and superfluid helium 4 is a complex number ψ. The magnitude of the complex number ψ is a fixed function of temperature, so the topological order parameter space is the set of complex numbers of magnitude $|\psi|$. Thus the order parameter space for superconductors and superfluids is a circle $\mathbb{S}^1$.

Example 4.1: Topological defects in nematic liquid crystals

Sethna problem 9.1

Example 4.2: XY defects

Sethna problem 9.2

Landau theory

Landau theory postulates that we can write down a function $\mathcal{L}$ known as the **Landau free energy**, or sometimes the **Landau functional**, which depends on the coupling constants K_i and the order parameter ϕ. $\mathcal{L}$ has the property that the state of the system is specified by the absolute minimum (i.e. global) of $\mathcal{L}$ with respect to ϕ. $\mathcal{L}$ has dimensions of energy, and is related to, but, as we will see, is not identical with the Gibbs free energy of the system. We assume that thermodynamic functions of state may be computed by differentiating $\mathcal{L}$, as if it were indeed the Gibbs free energy.

To specify $\mathcal{L}$, it is sufficient to use the following constraints on $\mathcal{L}$:

1. $\mathcal{L}$ has to consistent with the symmetries of the system.
2. Near T_c, $\mathcal{L}$ can be expanded in a power series in ϕ. i.e. $\mathcal{L}$ is an analytic function of both ϕ, and $[K]$. In a spatially uniform system of volume V, we can express the Landau free energy density as
3. In an inhomogeneous system with a spatially varying order parameter profile $\phi(\mathbf{r})$, $\mathcal{L}$ is a local function: i.e. it depends only on $\phi(\mathbf{r})$ and a finite number of derivatives.

4. In the disordered phase of the system, the order parameter $\phi = 0$, whilst it is small and non-zero in the ordered phase, near to the transition point. Thus, for $T > T_c$, $\phi = 0$ solves the minimum equation for $\mathcal{L}$; for $T < T_c$, $\phi \neq 0$ solves the minimum equation. Thus, for a homogeneous system,

where we have expanded $\mathcal{L}$ to $O(\phi^4)$ in the expectation that ϕ is small, and all the essential physics near T_c appears at this order. Whether or not the truncation of the power series for $\mathcal{L}$ is valid will turn out to depend on both the dimensionality of the system and the codimension of the singular point of interest.

Now we use the assumptions to explicitly construct $\mathcal{L}$.

For $T > T_c$, $\phi = 0$. Hence $a_1 = 0$. Now consider the symmetry constraint. This is best seen by example: for the ferromagnet $\phi = M$, and if we have an Ising system with $h = 0$, then the probability distribution P for ϕ is even in a finite system: $P(\phi) = P(-\phi)$. Assuming that $P \sim \exp(-\beta\mathcal{L})$, we require

The evenness of $\mathcal{L}$ in this case implies that,

Note that the requirement that $\mathcal{L}$ be analytic in ϕ precludes terms like $|\phi|$. Also, the coefficient $a_4(\{K_i\}) > 0$, otherwise the Landau function can be minimized by $\phi \to \infty$, whereas we wish to describe how the order parameter rises from zero and has a finite value as the coupling constants are varied through the transition point.

- $a_0(\{K_i\}, T)$ is simply the value of $\mathcal{L}$ in the high temperature phase, and we expect it to vary smoothly through T_c. It represents the degrees of freedom in the system which are not described by the order parameter, and so may be thought of as the smooth background, on which the singular behaviour is superimposed.
- Now consider a_4. Expanding in temperature near T_c, we obtain

It will be sufficient to just take a_4 to be a positive constant. The temperature dependence will turn out not to dominate the leading behavior of the thermodynamics near T_c.

- Similarly, we expand a_2:

We can determine the form of this from

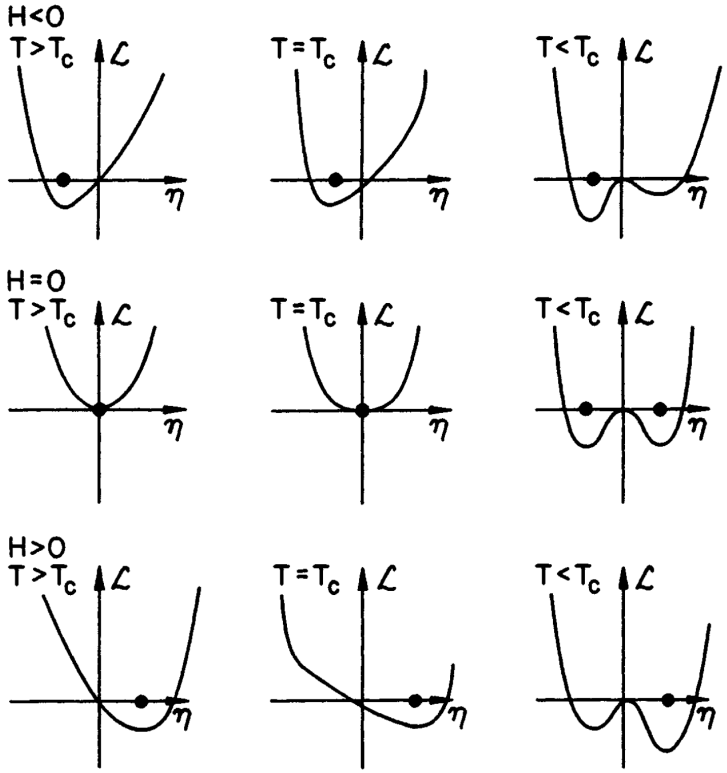
If ϕ is to be non-zero when $T < T_c$, then

Now let us extend the treatment to the case $h \neq 0$. For the Ising ferromagnet, we know that $\phi = M$ and that the appropriate energy is

In conclusion, we obtain our final expression

The coefficients a and b are phenomenological parameters, which in principle could be obtained from the appropriate microscopic theory. In principle, a term proportional to $h\phi^3$ is also allowed by symmetry, but this is not a leading term near the critical point.

The Landau free energy density for various values of T and h. The $\bullet$ indicates the value of ϕ at which $\mathcal{L}$ achieves its global minimum. The right-most column of graphs depicts the first order transition, which occurs for $T < T_c$ as h is varied from a negative to a positive value. The central row depicts the continuous transition, which occurs for $h = 0$ as T is varied from above T_c to below T_c.



Continuous phase transition

Here we discuss the continuous transition, which occurs for $h = 0$, as shown in the central row of graphs in figure. For $T > T_c$, the minimum of $\mathcal{L}$ is at $\phi = 0$. For $T = T_c$, the Landau function has zero curvature at $\phi = 0$, but $\phi = 0$ is still the global minimum. For $T < T_c$, two degenerate minima occur at $\phi = \pm\phi_s$. The value of ϕ_s depends on T:

The exponent β is found from the variation of the order parameter ϕ with t. For $t < 0$, we find that

enabling us to read off

For $t > 0$, $\mathcal{L} = 0$. For $t < 0$,

The heat capacity is computed from

This shows that the heat capacity exhibits a discontinuity and thus

To compute the remaining thermodynamic critical exponents, we need to let $h \neq 0$. Differentiating $\mathcal{L}$ with respect to h gives the magnetic equation of state for small ϕ:

On the critical isotherm, $t = 0$ and

which implies that

The isothermal susceptibility (recall that ϕ is the magnetization) is obtained by differentiating $\phi(h)$ with respect to h:

where $\phi(h)$ is a solution of

We are interested in the response function at zero external field. For $t > 0$, $\phi = 0$ and

For $t < 0$, $\phi^2 = -at/b$ and

Now we have found all of the mean field thermodynamic exponents. What about the critical exponents describing the spatial correlations in the system? To calculate these, we have to make an extension of Landau theory to deal with inhomogeneous systems. But first, while we are dealing with uniform systems, it is appropriate to mention the topic of first order phase transitions.

First order phase transitions

Landau theory is predicated on the assumption that the order parameter $\phi(t)$ is arbitrarily small as $t \to 0$. We saw that if

then for $h = 0$, $t \to 0$, phenomenological Landau theory gives a description of a continuous transition

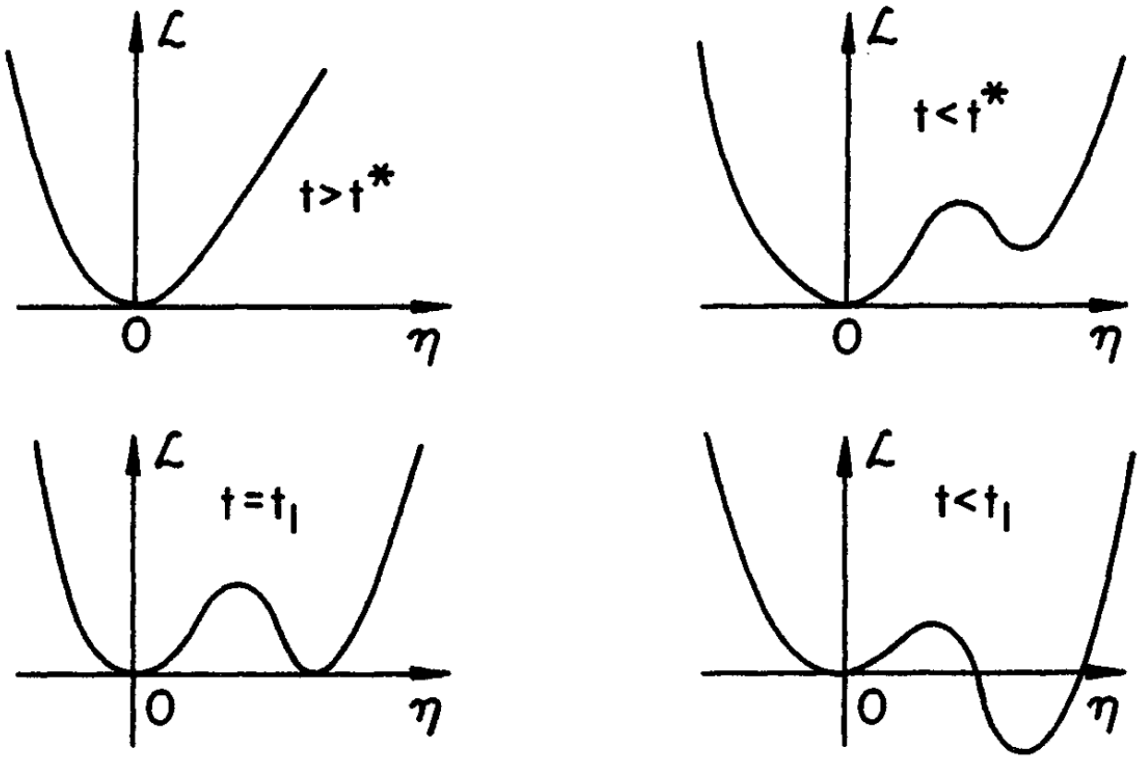
Now we consider more general expressions for $\mathcal{L}$.

In constructing $\mathcal{L}$, we saw that there cannot be a term linear in ϕ if the symmetric phase corresponds to $\phi = 0$. However, it was only the symmetry of the problem that prevented us from writing down a term cubic in ϕ. Let us now examine the effect of such a term, by considering

with a and b positive. Define $c \equiv 3C/4b$. Then, for $h = 0$, the equilibrium value of ϕ is obtained by differentiation with respect to ϕ and setting the result equal to zero:

The solution $\phi \neq 0$ becomes acceptable (i.e. real, not complex) for reduced temperatures t satisfying

i.e. $t < t_* = bc^2/a$. Since t_* is positive, this occurs at a temperature greater than T_c (T_c is the temperature where the coefficient of the term quadratic in ϕ vanishes). In the description of a continuous transition, we found that $\phi \neq 0$ only became acceptable for $t < 0$, i.e. $T < T_c$. We can see what is going on by sketching the form of $\mathcal{L}$. Without loss of generality, we choose $c < 0$.


$\mathcal{L}$
$t>t^*$
0
η
$\mathcal{L}$
$t<t^*$
0
η
$\mathcal{L}$
$t=t_1$
0
η
$\mathcal{L}$
$t<t_1$
0
η

For $t < t_*$, a secondary minimum and maximum have developed, in addition to the minimum at $\phi = 0$. As t is reduced further to the value t_1, the value of at the secondary minimum is equal to the value at $\phi = 0$. For $t < t_1$, the secondary minimum is now the global minimum, and the value of the order parameter which minimizes $\mathcal{L}$ jumps **discontinuously** from $\phi = 0$ to a non-zero value. This is a **first order transition**.

A general phenomenon associated with first-order transitions is the presence of **hysteresis** in cycling through the transition and the related effects of **superheating** and **supercooling**. Thermodynamically the transition should occur at t_1. However, the high temperature phase with $\phi = 0$ is stable against small fluctuations until the temperature t_1 is reached on cooling. Between t_1 and $t = 0$ (or $T = T_c$), the ordered phase ($\phi \neq 0$) will only occur if there is a sufficiently large fluctuation. Below $t = 0$ the system is unstable against infinitesimal fluctuations into the ordered state. Thus, on cooling, the actual transition to the low-temperature ordered phase will occur at some temperature between t_1 and $t = 0$ depending on the particular sample and the experimental conditions. This is the **supercooled** phase. Similar arguments show that, on heating, the transition from the ordered to the high temperature phase will occur between t_1 and t_*. This is the **superheated** phase.

First order phase transition: In this case, we find that

Note that at the first order transition, $\phi(t_1)$ is not arbitrarily small as $t \to t_1^-$, so Landau theory is not valid. Thus, if there is no symmetry reason that forces $C = 0$, then the cubic term will in general cause a first order transition to occur. In more general situations, where the order parameter has more than one component, then $\mathcal{L}$ —a scalar—is constructed out of combinations of the order parameter components which are invariants under the symmetries of the physical situation. Then, a sufficient, but not necessary condition for the existence of a continuous transition is that there are no cubic invariants consistent with the symmetry of the problem.

The Landau criterion is, of course, only a statement made within the context of mean field theory. It is important to realize that the fluctuation effects omitted by mean field theory can change the order of a transition.

Inhomogeneous system

To obtain the mean field value of the critical exponents ν and η, which describes behavior of the correlation function, we invoke the **Ornstein-Zemike** extension to Landau theory. We allow the order parameter, in this case the magnetization, to vary with position.

We write the Landau free energy as

Go to Fourier space

Since $\vec{m}(\mathbf{r})$ is real, we get

Similarly,

In all, we get

We see that all the modes are decoupled. By the classical equipartition theorem,

The spin-spin correlation function is

Take the Fourier transform

Assume we are in the paramagnetic phase, so

Since the modes are uncorrelated,

Take the inverse Fourier transform,

ξ is the correlation length.

This means that

Validity of mean field theory

If we compare to the critical exponents, we see that they do not quite agree.

Table 3.1. Universality classes

Universality class	Symmetry of order parameter	α	β	γ	δ	ν	η	Physical examples
2-d Ising	2-component scalar	0 (log)	1/8	7/4	15	1	1/4	some adsorbed mono e.g. H on Fe
3-d Ising	2-component scalar	0.10	0.33	1.24	4.8	0.63	0.04	phase separation, flu order-disorder e.g. β
3-d X-Y	2-dimensional vector	0.01	0.34	1.30	4.8	0.66	0.04	superfluids, supercon
3-d Heisenberg	3-dimensional vector	−0.12	0.36	1.39	4.8	0.71	0.04	isotropic magnets
mean-field		0 (dis.)	1/2	1	3	1/2	0	
2-d Potts, q=3	q-component scalar	1/3	1/9	13/9	14	5/6	4/15	some adsorbed mono
q=4		2/3	1/12	7/6	15	2/3	1/4	e.g. Kr on graphite

Mean field theory can only be valid when fluctuations are unimportant. Consider spin alignment in d-dimensions. It is intuitively reasonable that mean field theory, where the interaction is with an average of all the other spins in the system, should improve as the number of nearest neighbors or the range of the interaction increases. It can be proven that mean field theory become exact as the number of interacting neighbors z, the dimensionality, or the range of interactions (if limit is taken carefully) become infinite.

The energy of typical fluctuation is of the order $k_B T$ and its size, is determined by the correlation length ξ, is of order ξ^d. The contribution to the free energy per unit volume is

The specific heat is

Integrating twice, we find

For a consistent theory, the free energy of the fluctuation must be much less than the total free energy. This implies that as $t \to 0$,

This a simple case of the **Ginzburg criterion**.

Hence, mean field theory is consistent and the exponents are correct if $d > 4$. $d = 4$ is the **upper critical dimension**. Different Landau expansions lead to different critical exponents and hence different upper critical dimensions. All model with the same Landau expansion share the same critical exponents. Hence mean field theory leads to **universality classes**, although the degree of universality is overstated.

Summary

Mean field theory usually gives correct qualitative predictions for the phase diagrams of three-dimensional systems. It provides a feasible approach to complicated problems (often the only one), and it can be used as a starting point for more sophisticated calculations. Whereas many numerical approaches get harder as the dimensionality increase, mean field theory improves. Finally, it highlights the importance of the symmetry of the order parameter.

Example 4.3: Symmetries and wave equations

Sethna 9.6 (Symmetries and wave equations)

Example 4.4: Superfluid order and vortices

Sethna 9.7 (Superfluid order and vortices)

Extra Notes